

PE-EC701B — Satellite Communication

Semester VII • 3L:0T:0P • 3 Credits

Module 4 Exam Notes: Typical Phenomena in Satellite Communication

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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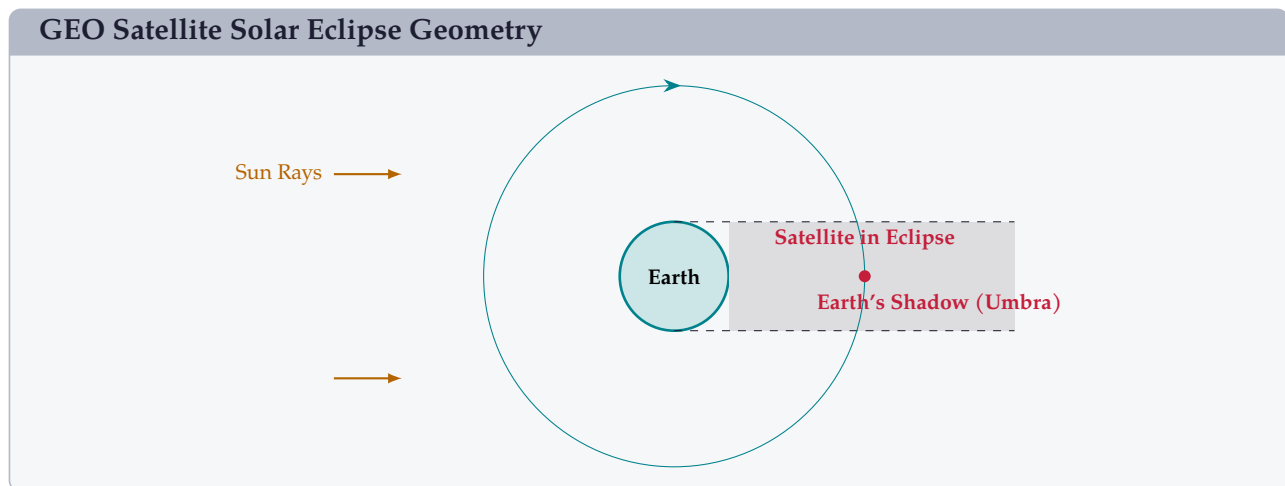
1. Solar Eclipse on Satellite

An orbital eclipse occurs when a satellite passes through the shadow cast by the Earth (or occasionally the Moon), blocking its direct line of sight to the Sun.

Solar Eclipse: The condition where Earth's shadow blocks sunlight from reaching the satellite's solar panels. For geostationary (GEO) satellites, eclipses occur twice a year during the vernal and autumnal equinoxes for a 72-day period.

1.1 Eclipse Geometry and Occurrence

Because the Earth's axis of rotation is tilted at 23.4° , GEO satellites bypass Earth's shadow during most of the year. However, during the equinoxes (late March and late September), the Sun lies in the Earth's equatorial plane, aligning the Sun, Earth, and GEO satellites.



1.2 Effects of Eclipse

- **Loss of Power Generation:** The solar panels cease to generate electricity, causing the bus voltage to drop unless batteries take over.
- **Thermal Shock:** The temperature drops rapidly from over $+100^\circ\text{C}$ in sunlight to below -100°C in shadow, placing mechanical stress on components.

1.3 Remedies

- **Secondary Battery Backup:** Lithium-Ion or Ni-H₂ batteries power the payload during the eclipse.
- **Electric Thermal Heaters:** Automatically switch on to keep sensitive transponder components at safe operating temperatures.

2. Sun Transit Outage

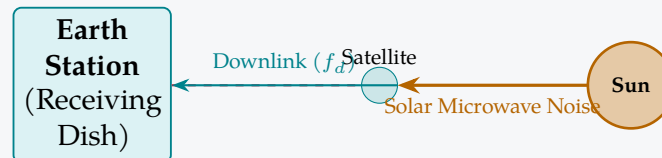
Sun Transit Outage is a natural phenomenon that causes temporary signal degradation when the Sun passes directly behind the satellite as seen from the receiving Earth station.

Sun Transit Outage: A downlink interference event occurring when the Sun, the satellite, and the ground receiving dish align in a straight line. The Sun's intense thermal microwave emissions overwhelm the satellite's downlink signal.

2.1 Alignment Geometry

During this alignment, the ground station antenna points directly at the satellite and, simultaneously, directly at the Sun.

Sun Transit Outage Alignment



- **Effects:** The Sun acts as an extremely high-temperature noise source ($T_{sun} > 10,000$ K at microwave frequencies). The antenna noise temperature (T_{ant}) spikes, reducing the receiver's carrier-to-noise ratio (C/N) to near zero, resulting in a complete communications dropout lasting up to 10 minutes.
- **Remedies:** Sun outages cannot be prevented physically. Ground operators mitigate the impact by:
 - Rerouting critical traffic to adjacent satellites.
 - Using dual-antenna setups pointed at different orbital slots.

3. Doppler Frequency Shift

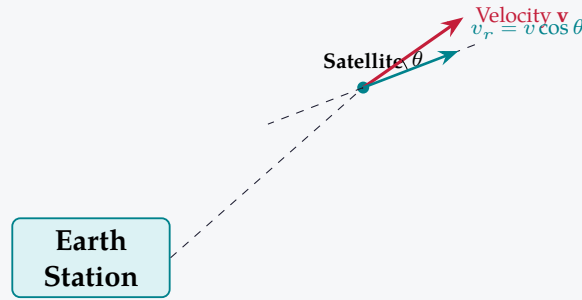
When a satellite moves relative to a terrestrial Earth station, the received frequency shifts from the transmitted frequency due to the Doppler effect.

Doppler Shift: The change in apparent frequency of a received electromagnetic wave caused by relative motion between the transmitter (satellite) and receiver (ground station).

3.1 Mathematical Expression for Doppler Shift

Let a satellite move with velocity vector $\mathbf{v}$ making an angle θ relative to the line of sight (LOS) to the Earth station.

Doppler Vector Components



The relative radial velocity (v_r) along the line of sight is:

$$v_r = v \cos(\theta) \quad (1)$$

If f is the transmitted frequency and c is the speed of light, the received frequency f' is:

$$f' = f \left(1 \pm \frac{v_r}{c} \right) \quad (2)$$

The Doppler frequency shift (Δf) is defined as:

$$\Delta f = f' - f = \pm f \frac{v_r}{c} = \pm \frac{v \cos(\theta)}{\lambda} \quad (3)$$

Where:

- Δf = Doppler frequency shift (Hz)
- f = Transmitted carrier frequency (Hz)
- $v_r = v \cos(\theta)$ = Radial velocity component (m/s)
- $c \approx 3 \times 10^8$ m/s = Speed of light
- The sign is **positive (+)** when the satellite is approaching the station.
- The sign is **negative (-)** when the satellite is receding from the station.

3.2 Impact on Satcom Systems

1. **High Frequency Vulnerability:** Since $\Delta f \propto f$, Doppler shift is severe at high frequencies (Ku and Ka bands) and in Low Earth Orbit (LEO) systems where orbital speed is extremely high ($v \approx 7.5$ km/s).
2. **Receiver Tracking:** Receiver circuits must implement automatic frequency control (AFC) loops to track the frequency drift and maintain demodulator lock.

Quick Revision — Module IV

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Key Formulas at a Glance

Doppler Frequency Shift	$\Delta f = \pm f \frac{v_r}{c} = \pm \frac{v_r}{\lambda}$
Relative Radial Velocity	$v_r = v \cos(\theta)$
Maximum Eclipse Duration	$t_{max} \approx 72$ minutes (occurs at Equinox center)
Doppler Shift (Approaching)	$f_{received} = f_{transmitted} \left(1 + \frac{v_r}{c}\right)$

One-Line Definitions

- **Solar Eclipse:** Temporary blockage of sunlight to a satellite by Earth's shadow.
- **Equinox:** The twice-yearly period (March/September) when solar eclipses occur for GEO satellites.
- **Sun Transit Outage:** Alignment of ground station, satellite, and Sun, causing thermal noise outages.
- **Doppler Shift:** Apparent frequency deviation of a carrier wave due to relative motion.
- **Radial Velocity (v_r):** The component of satellite velocity directly along the ground station line of sight.
- **Umbra:** The central, completely dark portion of Earth's shadow.
- **AFC:** Automatic Frequency Control, used to dynamically track Doppler carrier shifts.

Frequently Asked / Viva Questions

Q1. Why do GEO satellites experience eclipses only during equinoxes?

Because the Earth's orbital plane is tilted at 23.4° . Eclipses occur only when the Sun crosses the equator (during spring and autumn equinoxes), aligning the Sun, Earth, and satellite.

Q2. What are the major effects of a solar eclipse on a satellite?

Complete loss of primary solar power generation (requiring battery operation) and severe thermal shock (temperatures drop from $+100^\circ\text{C}$ to -100°C).

Q3. How does a satellite thermal control system mitigate eclipse effects?

It activates electric heaters to maintain minimum operating temperatures for sensitive electronics and uses multi-layer insulation (MLI) to retain heat.

Frequently Asked / Viva Questions (contd.)

Q4. Explain the physical cause of a Sun Transit Outage.

The Sun is a massive blackbody radiator emitting intense microwave noise. When aligned behind the satellite, this thermal noise fills the ground receiver dish beam, blocking the satellite downlink.

Q5. State the duration and annual schedule of Sun Transit Outages.

They occur twice a year (near equinoxes) for a few consecutive days, lasting about 2 to 10 minutes per day.

Q6. Derive the mathematical expression for Doppler frequency shift.

Let a source emit waves of period T . Due to radial motion v_r , the wave travels distance $v_r T$ per cycle. The change in apparent wavelength is $\Delta\lambda = \pm v_r / f$. Since $f = c/\lambda$, differentiating yields $\Delta f = \pm f(v_r/c)$.

Q7. Under what conditions is Doppler shift maximum and zero?

Doppler shift is maximum when $\theta = 0^\circ$ (satellite is moving directly toward or away from the station). It is zero when $\theta = 90^\circ$ (satellite motion is perpendicular to the line of sight).

Q8. Why is Doppler shift more severe in LEO satellites than in GEO satellites?

LEO satellites travel at extremely high speeds (7.5 km/s) relative to Earth, creating huge radial velocity changes. GEO satellites are relatively stationary, experiencing negligible Doppler shift.

Q9. Calculate the maximum Doppler shift for a LEO satellite ($v_r = 7$ km/s) transmitting at 10 GHz.

$$\Delta f = f \cdot (v_r/c) = 10^{10} \times (7 \times 10^3)/(3 \times 10^8) = 7 \times 10^{13}/(3 \times 10^8) \approx 233.33 \text{ kHz.}$$

Q10. What is the difference between Umbra and Penumbra?

Umbra is the inner region of the shadow where the light source is completely blocked. Penumbra is the outer region where only a portion of the light source is blocked.

Q11. Why are Sun outages independent of the satellite's power status?

Because they are caused by external solar noise interfering with the ground receiver, not by any power loss on the satellite itself.

Q12. How does a ground receiver maintain lock during a high Doppler LEO pass?

By using a phase-locked loop (PLL) with wide tracking bandwidth or digital frequency estimation algorithms to dynamically adjust the receiver local oscillator.

Comparison of Solar Eclipse and Sun Transit Outage

Parameter	Solar Eclipse	Sun Transit Outage
Affected Unit	Satellite (Spacecraft)	Ground Station (Earth Terminal)
Physical Cause	Earth blocks solar light to the satellite.	Sun aligns behind the satellite.
Primary Effect	Loss of solar power, temperature drop.	Severe thermal noise, downlink carrier loss.
System Remedy	Secondary batteries, thermal heaters.	Traffic rerouting, antenna diversity.
Max Duration	Up to 72 minutes per day.	Up to 10 minutes per day.